```
import pandas as pd
import numpy as np

df = pd.read_csv("Advanced_pandas_sales_dataset.csv")
df.head()
```

	Order_ID	Order_Date	Customer	City	Region	Product	Category	Sales	Quantity	Discount	Profit
0	1001	2024-10-01	Priya	Hyderabad	East	Mouse	Accessories	1735	7	0.00	307.52
1	1002	2024-05-03	Sneha	Kolkata	East	Monitor	Electronics	807	2	0.15	49.89
2	1003	2024-01-26	Anita	Mumbai	West	Phone	Electronics	2328	1	0.20	66.47
3	1004	2024-06-06	Sara	Pune	East	Keyboard	Accessories	3933	3	0.20	-71.66
4	1005	2024-10-15	Rahul	Bangalore	South	Monitor	Electronics	3550	2	0.20	146.30

Q1. Dataset Exploration

Perform dataset exploration by displaying the first 10 rows, dataset structure, shape, and data types.

```
df.head(10)
df.info()
df.shape
df.dtypes
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 230 entries, 0 to 229
Data columns (total 11 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Order_ID    230 non-null    int64
 1   Order_Date  230 non-null    object
 2   Customer    230 non-null    object
 3   City        230 non-null    object
 4   Region      230 non-null    object
 5   Product     230 non-null    object
 6   Category    230 non-null    object
 7   Sales       230 non-null    int64
 8   Quantity    230 non-null    int64
 9   Discount    230 non-null    float64
 10  Profit      230 non-null    float64
dtypes: float64(2), int64(3), object(6)
memory usage: 19.9+ KB
```

	0
Order_ID	int64
Order_Date	object
Customer	object
City	object
Region	object
Product	object
Category	object
Sales	int64
Quantity	int64
Discount	float64
Profit	float64

dtype: object

Explanation

This step is essential to understand the dataset before analysis. The head() function helps in viewing sample data, while info() provides details about columns, non-null values, and data types. The shape shows the number of rows and columns, and dtypes identifies whether the data is numerical or categorical. This understanding helps in selecting appropriate data processing and analysis techniques.

Q2. Sales Filtering Analysis

Filter the dataset to display:

Orders where Sales > 3500 Orders where Region = "East" and Sales > 2500

```
df[df["Sales"] > 3500]
```

```
df[(df["Region"] == "East") & (df["Sales"] > 2500)]
```

	Order_ID	Order_Date	Customer	City	Region	Product	Category	Sales	Quantity	Discount	Profit
3	1004	2024-06-06	Sara	Pune	East	Keyboard	Accessories	3933	3	0.20	-71.66
7	1008	2024-03-18	Kavya	Kolkata	East	Phone	Electronics	4693	7	0.05	918.49
8	1009	2024-10-31	Kavya	Ahmedabad	East	Phone	Electronics	5071	6	0.10	707.55
13	1014	2024-03-18	Anita	Bangalore	East	Mouse	Accessories	3616	4	0.05	463.93
21	1022	2024-01-14	Nitin	Chennai	East	Monitor	Electronics	4827	6	0.05	912.34
27	1028	2024-07-21	Dev	Kolkata	East	Tablet	Electronics	4054	6	0.00	1013.83
33	1034	2024-12-15	Nitin	Ahmedabad	East	Tablet	Electronics	4378	2	0.20	-104.30
34	1035	2024-02-20	Nitin	Ahmedabad	East	Laptop	Electronics	2535	4	0.10	188.15
41	1042	2024-07-15	Rohan	Chennai	East	Laptop	Electronics	3268	8	0.15	295.15
43	1044	2024-09-20	Kavya	Hyderabad	East	Printer	Office	2618	7	0.05	454.64
52	1053	2024-05-24	Anita	Chennai	East	Phone	Electronics	5949	7	0.20	220.44
55	1056	2024-01-03	Sneha	Chennai	East	Headphones	Accessories	2784	4	0.05	474.60
58	1059	2024-12-30	Anita	Delhi	East	Tablet	Electronics	4790	1	0.20	238.53
60	1061	2024-02-05	Nitin	Delhi	East	Monitor	Electronics	4508	5	0.00	972.35
70	1071	2024-05-28	Vikas	Hyderabad	East	Mouse	Accessories	3762	2	0.05	719.68
71	1072	2024-05-20	Rahul	Chennai	East	Printer	Office	3696	2	0.10	600.75
73	1074	2024-10-10	Sara	Bangalore	East	Mouse	Accessories	4004	1	0.15	445.81
74	1075	2024-11-25	Pooja	Ahmedabad	East	Laptop	Electronics	3865	8	0.20	174.96
76	1077	2024-03-02	Neha	Pune	East	Monitor	Electronics	2964	8	0.20	109.33
81	1082	2024-08-16	Isha	Chennai	East	Phone	Electronics	2720	4	0.15	163.13
87	1088	2024-07-29	Pooja	Mumbai	East	Headphones	Accessories	4811	4	0.20	-25.47
92	1093	2024-01-13	Karan	Kolkata	East	Laptop	Electronics	5372	3	0.15	136.64
96	1097	2024-06-07	Isha	Delhi	East	Phone	Electronics	5096	4	0.15	283.66
98	1099	2024-07-26	Rahul	Kolkata	East	Laptop	Electronics	5674	7	0.05	1141.51
108	1109	2024-10-23	Raj	Hyderabad	East	Tablet	Electronics	2509	4	0.05	371.67
109	1110	2024-11-28	Priya	Ahmedabad	East	Printer	Office	2561	4	0.20	57.96
112	1113	2024-11-01	Meera	Bangalore	East	Tablet	Electronics	4179	5	0.00	754.64
113	1114	2024-05-10	Rahul	Bangalore	East	Mouse	Accessories	3285	3	0.00	625.46
116	1117	2024-12-07	Pooja	Kolkata	East	Tablet	Electronics	3758	5	0.15	446.54
128	1129	2024-06-09	Nitin	Delhi	East	Laptop	Electronics	5713	3	0.00	1498.02
139	1140	2024-05-04	Karan	Ahmedabad	East	Monitor	Electronics	3500	8	0.05	467.16
141	1142	2024-01-01	Rahul	Mumbai	East	Headphones	Accessories	5430	8	0.05	741.25
146	1147	2024-05-31	Raj	Delhi	East	Keyboard	Accessories	5709	7	0.20	383.58
152	1153	2024-02-27	Rahul	Mumbai	East	Phone	Electronics	3644	2	0.05	512.25
156	1157	2024-05-27	Neha	Kolkata	East	Laptop	Electronics	4906	4	0.00	1115.89
171	1172	2024-09-22	Neha	Hyderabad	East	Headphones	Accessories	4228	1	0.20	236.15
179	1180	2024-12-13	Neha	Ahmedabad	East	Monitor	Electronics	2690	7	0.10	452.70
182	1183	2024-12-01	Pooja	Chennai	East	Laptop	Electronics	5884	1	0.05	1266.10
185	1186	2024-03-12	Kavya	Ahmedabad	East	Phone	Electronics	5047	2	0.10	564.78
187	1188	2024-07-10	Isha	Kolkata	East	Mouse	Accessories	4333	8	0.10	667.53
193	1194	2024-07-23	Pooja	Delhi	East	Laptop	Electronics	3767	7	0.20	103.23
197	1198	2024-10-16	Rahul	Ahmedabad	East	Tablet	Electronics	3955	1	0.10	284.92
202	1203	2024-05-10	Rahul	Mumbai	East	Phone	Electronics	2901	8	0.15	252.25
207	1208	2024-04-29	Dev	Chennai	East	Monitor	Electronics	3585	7	0.15	287.44
216	1217	2024-08-16	Karan	Chennai	East	Monitor	Electronics	2643	5	0.15	78.53
223	1224	2024-01-07	Dev	Bangalore	East	Tablet	Electronics	4115	4	0.05	870.03
225	1226	2024-06-14	Anita	Ahmedabad	East	Laptop	Electronics	5647	2	0.15	662.00
228	1229	2024-09-04	Priya	Pune	East	Tablet	Electronics	2635	8	0.00	531.76

Explanation

Filtering helps in identifying high-value transactions. By selecting records where sales exceed a certain threshold, we can focus on important revenue-generating transactions. Combining multiple conditions (Region and Sales) provides deeper insights, such as identifying high-performing regions. This is useful for decision-making and targeted strategies.

Q3. Customer Purchase Analysis

Identify the top 5 customers based on total sales.

```
df.groupby("Customer")["Sales"].sum().sort_values(ascending=False).head(5)
```

Customer	Sales
Nitin	59820
Raj	58416
Rahul	52291
Dev	48500
Isha	46818

dtype: int64

Explanation

Grouping data by customer and summing sales allows us to evaluate customer contribution. Sorting in descending order helps identify top customers who generate the highest revenue. These customers are valuable for the business and can be targeted for retention strategies, loyalty programs, and personalized marketing.

Q4. Regional Sales Performance

Calculate the total sales generated by each region and identify the region with the highest revenue.

```
region_sales = df.groupby("Region")["Sales"].sum()

region_sales

region_sales.idxmax()
```

```
'East'
```

Explanation

This analysis helps in understanding geographical performance. By aggregating sales region-wise, we can identify which regions contribute most to revenue. The idxmax() function helps in finding the top-performing region. Businesses can use this insight to allocate resources and expand operations in high-performing areas

Q5. Product Performance Analysis

Calculate the total quantity sold for each product and identify the most popular product.

```
product_qty = df.groupby("Product")["Quantity"].sum()

product_qty

product_qty.idxmax()
```

```
'Monitor'
```

Explanation

This step focuses on product demand. By summing quantity sold per product, we can identify which products are most popular among customers. This helps businesses in inventory planning, marketing focus, and production decisions.

Q6. Multi-Level Grouping

Group the dataset by Region and Category and calculate total sales for each combination.

```
df.groupby(["Region", "Category"])["Sales"].sum()
```

Region	Category	Sales
East	Accessories	57646
	Electronics	160907
	Office	14122
North	Accessories	62052
	Electronics	70023
	Office	26135
South	Accessories	80914
	Electronics	86690
	Office	15504
West	Accessories	64811
	Electronics	84128
	Office	21288

dtype: int64

Explanation

Multi-level grouping provides a deeper level of analysis by combining multiple variables. It helps in understanding how different product categories perform across regions. This insight can guide targeted marketing strategies and region-specific product offerings.

Q7. Pivot Table Creation

Create a pivot table showing:

Rows → Region Columns → Product Values → Total Sales

```
pivot = pd.pivot_table(df, values="Sales", index="Region", columns="Product", aggfunc="sum")

pivot
```

Product	Headphones	Keyboard	Laptop	Monitor	Mouse	Phone	Printer	Tablet
Region								
East	23899	10845	55568	31299	22902	39667	14122	34373
North	15450	20538	4540	14922	26064	30535	26135	20026
South	24111	23480	23029	19167	33323	18218	15504	26276
West	17159	38499	10826	18178	9153	27850	21288	27274

Explanation

Pivot tables transform data into a structured format, making it easier to compare values across categories. This helps in identifying trends, such as which products perform well in specific regions. It simplifies complex data analysis.

Q8. Profitability Analysis

Calculate the average profit for each product category and identify the most profitable category.

```
profit_cat = df.groupby("Category")["Profit"].mean()

profit_cat

profit_cat.idxmax()
```

```
'Office'
```

Explanation

Profitability analysis helps in understanding which categories generate the most profit on average. This is important for pricing strategies, cost control, and business focus. High-profit categories can be prioritized for growth.

Q9. Discount Impact Study

Analyze the relationship between discount and profit by calculating average profit for each discount level.

```
df.groupby("Discount")["Profit"].mean()
```

Discount	Profit
0.00	751.210588
0.05	531.368039
0.10	346.460513
0.15	237.191400
0.20	90.026923

dtype: float64

Explanation

This analysis helps in understanding how discounts affect profitability. If higher discounts lead to lower profit, it indicates that excessive discounting may harm business margins. This insight is useful for designing pricing strategies.

Q10. End-to-End Data Wrangling Task

Perform the following:

Filter dataset where Sales > 3000 Group by Region and Product Calculate total sales Convert into pivot table

```
filtered = df[df["Sales"] > 3000]

grouped = filtered.groupby(["Region", "Product"])["Sales"].sum()

pivot_final = grouped.unstack()

pivot_final
```

Product	Headphones	Keyboard	Laptop	Monitor	Mouse	Phone	Printer	Tablet
Region								
East	14469	9642	44096	16420	19000	29500	3696	29229
North	5326	15653	4540	5087	22209	24670	17185	3392
South	10026	19249	17950	14921	26928	18218	14143	19613
West	13898	33380	5922	14354	3511	15653	12648	17185

Explanation

This is a complete data analysis workflow combining filtering, grouping, and reshaping. It focuses on high-value transactions and presents them in a structured format. This helps in identifying top-performing products in each region and supports strategic decision-making.

```
Start coding or generate with AI.
```